

Electric Circuits

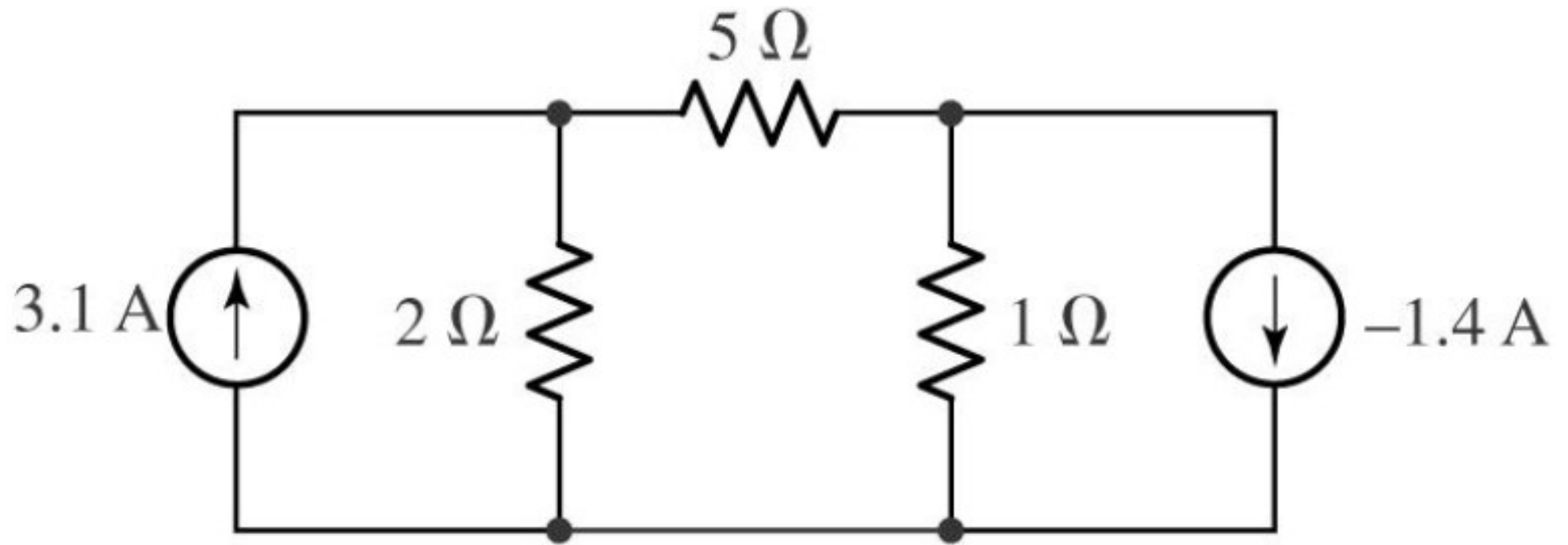


*Nodal Analysis: Circuits with Independent
& Dependent Current Sources*

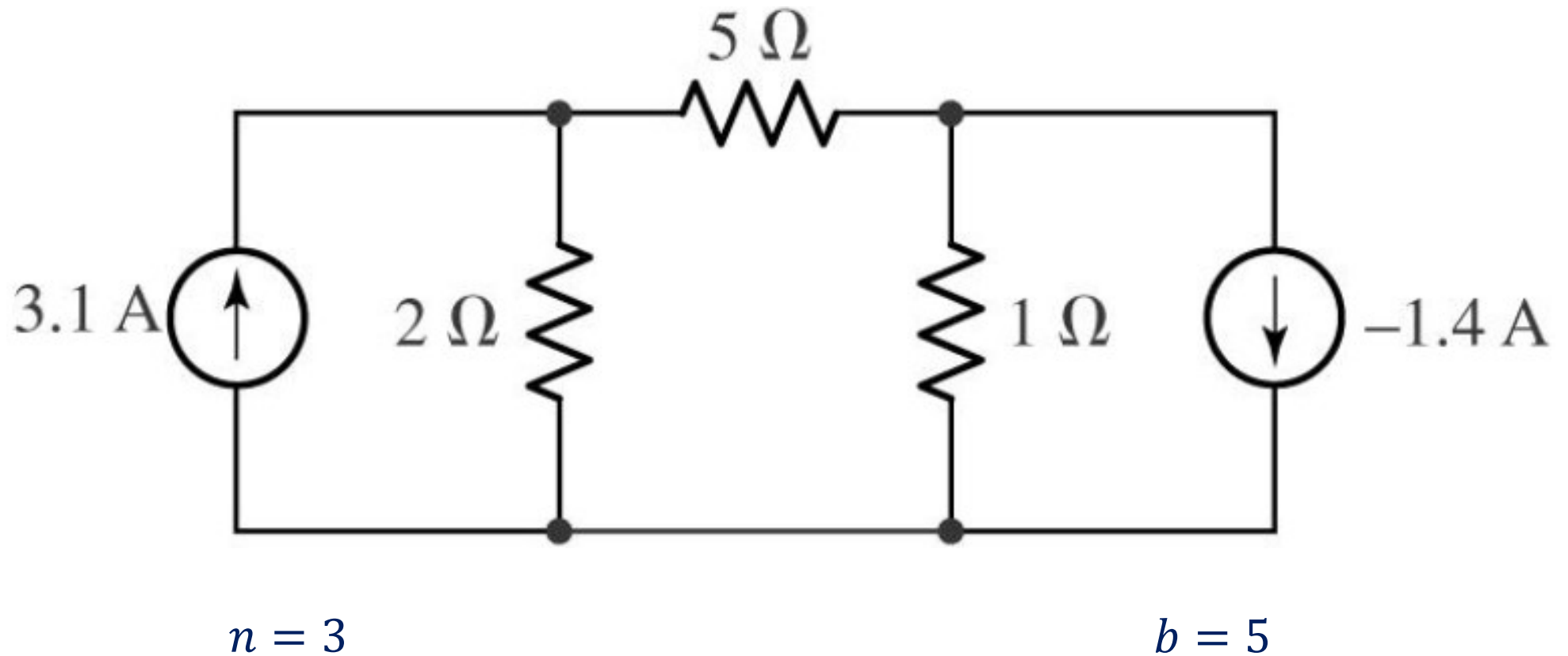
Tentative Lecture Plan

L	LDay-1	LDay-2	Topics	T	TDay
1	FRI, 04 AUG	MON, 07 AUG	Basic Definitions, Voltage & Current Sources, KCL & KVL	1	THU, 10 AUG
2	TUE, 08 AUG	FRI, 11 AUG	Nodal Analysis: Independent and Dependent Current Sources	2	SAT, 19 AUG
3	MON, 14 AUG	THU, 17 AUG	Nodal Analysis: Current and Voltage Sources		
4	FRI, 18 AUG	MON, 21 AUG	Mesh Analysis: Voltage and Current Sources	3	THU, 24 AUG
5	TUE, 22 AUG	FRI, 25 AUG	Network Theorems: Superposition Theorem		
SUN, 27 AUG : Quiz-1 : Lectures 1-5					
6	FRI, 26 AUG	MON, 28 AUG	Network Theorems: Thevenin's and Norton's Theorem	4	THU, 31 AUG
7	TUE, 29 AUG	FRI, 04 SEP	Network Theorems: Maximum Power Transfer Theorem	5	WED, 06 SEP
8	TUE, 05 SEP	FRI, 08 SEP	Series RL and RC Circuits		
9	SAT, 09 SEP	MON, 11 SEP	Series RL, RC, RLC Circuits Response to Constant, Sinusoid and Exponential Input	6	THU, 14 SEP
10	TUE, 12 SEP	FRI, 15 SEP	Concept of Phasor, AC Circuits, AC Power	R	SAT, 16 SEP
WED, 20 SEP: MidSem : Lectures 1 – 10					

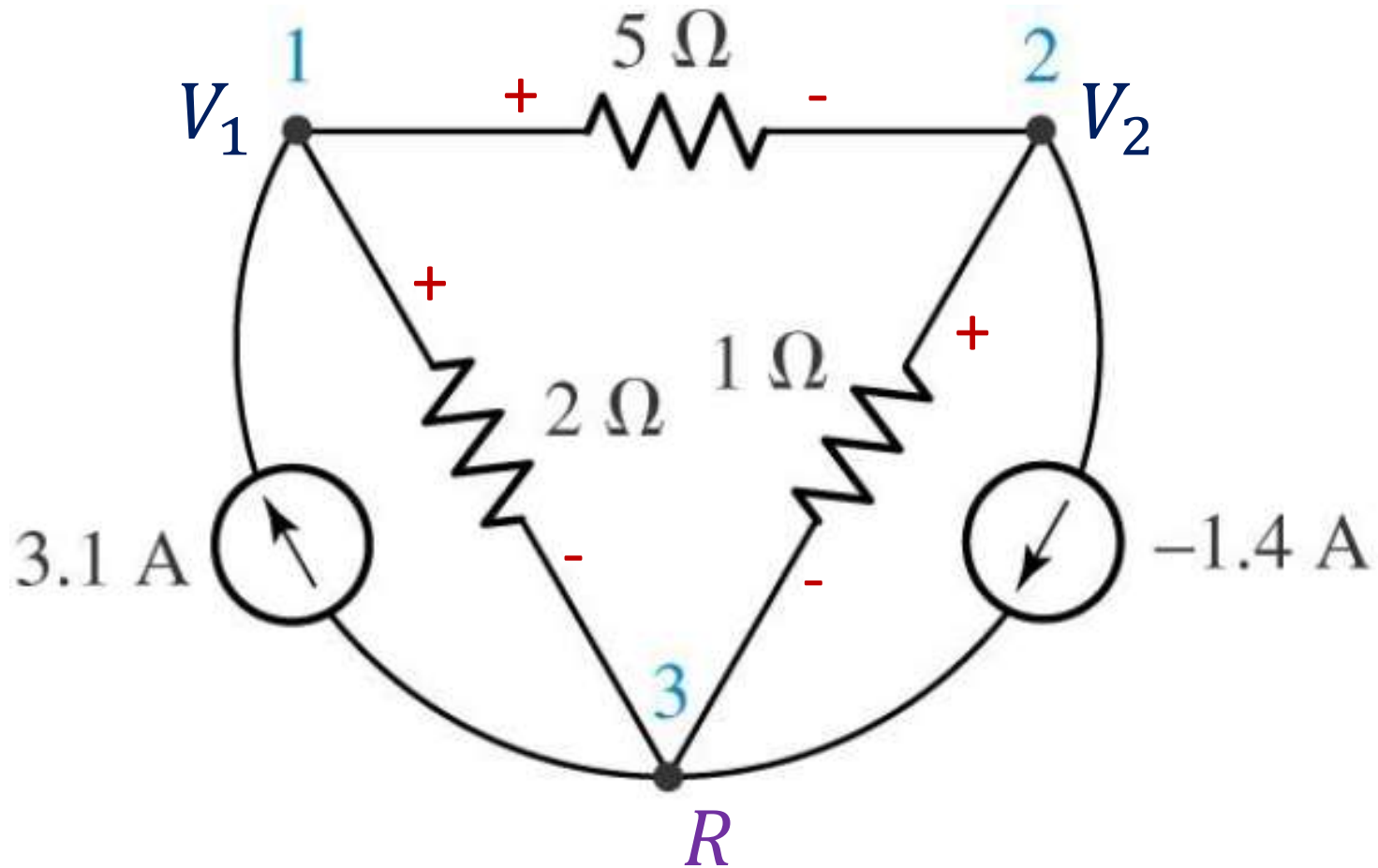
Analyzing A Simple Circuit



Analyzing A Simple Circuit



Analyzing A Simple Circuit



Analyzing A Simple Circuit

- KCL at Node **1**

- $-\frac{V_1}{2} - \left(\frac{V_1 - V_2}{5}\right) + 3.1 = 0$

- KCL at Node **2**

- $\frac{V_1 - V_2}{5} - \frac{V_2}{1} - (-1.4) = 0$

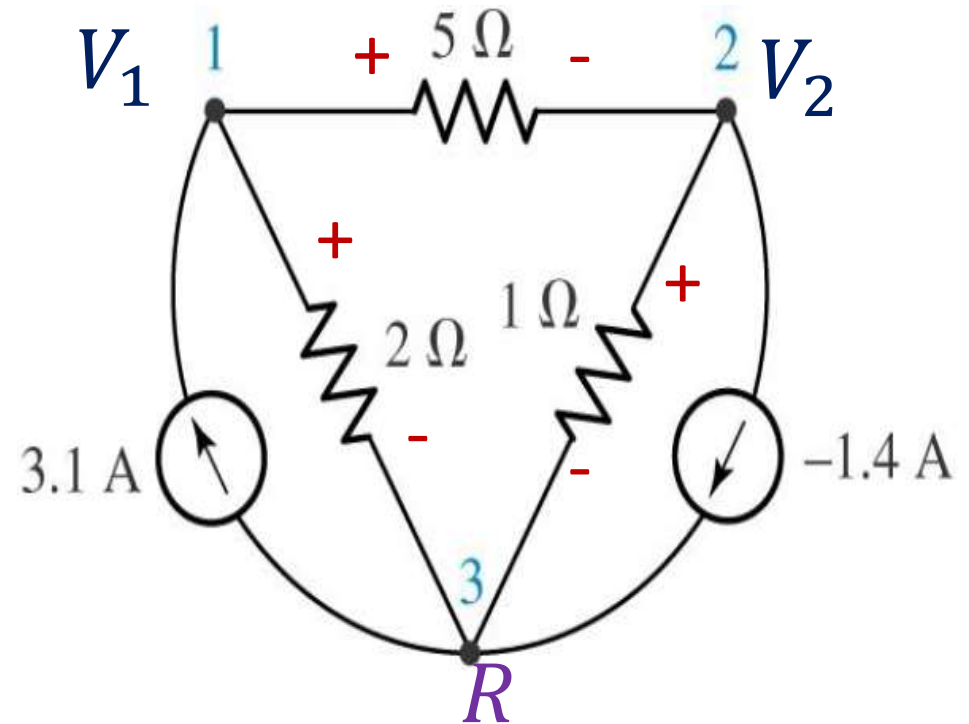


$$7V_1 - 2V_2 = 31$$

$$V_1 - 6V_2 = -7$$

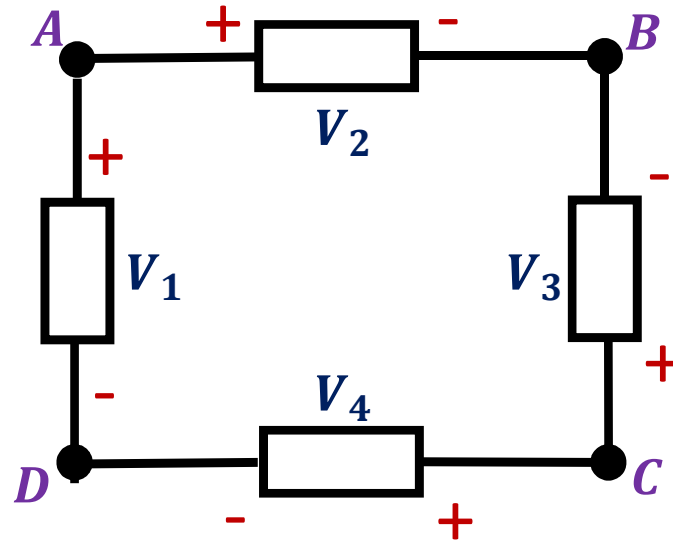
$$V_1 = 5V$$

$$V_2 = 2V$$



Kirchoff's Voltage Law (KVL)

KVL: The algebraic sum of all voltages between successive nodes in a closed path in the circuit is equal to zero

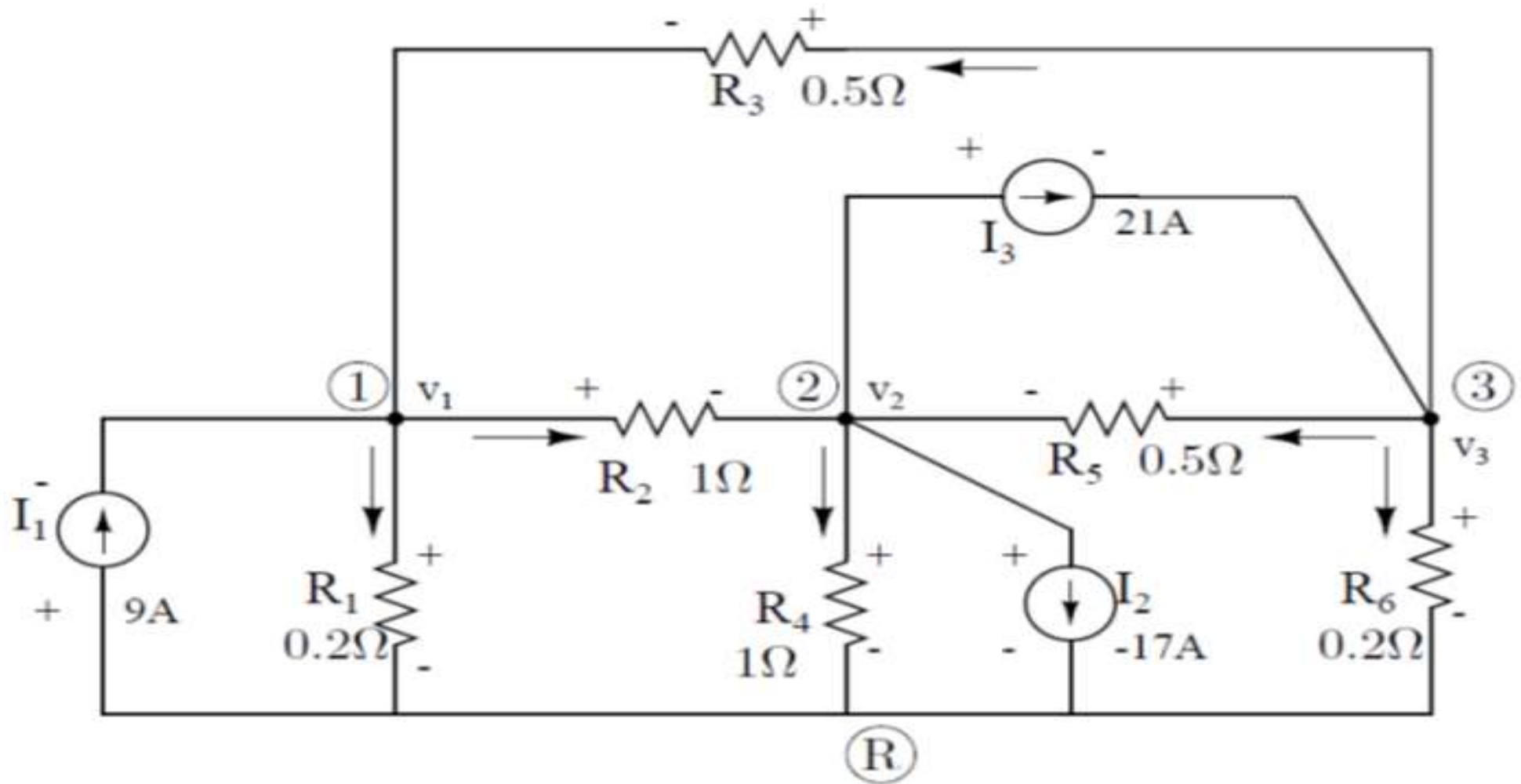


$$A - B - C - D - A$$
$$V_2 - V_3 + V_4 - V_1 = 0$$

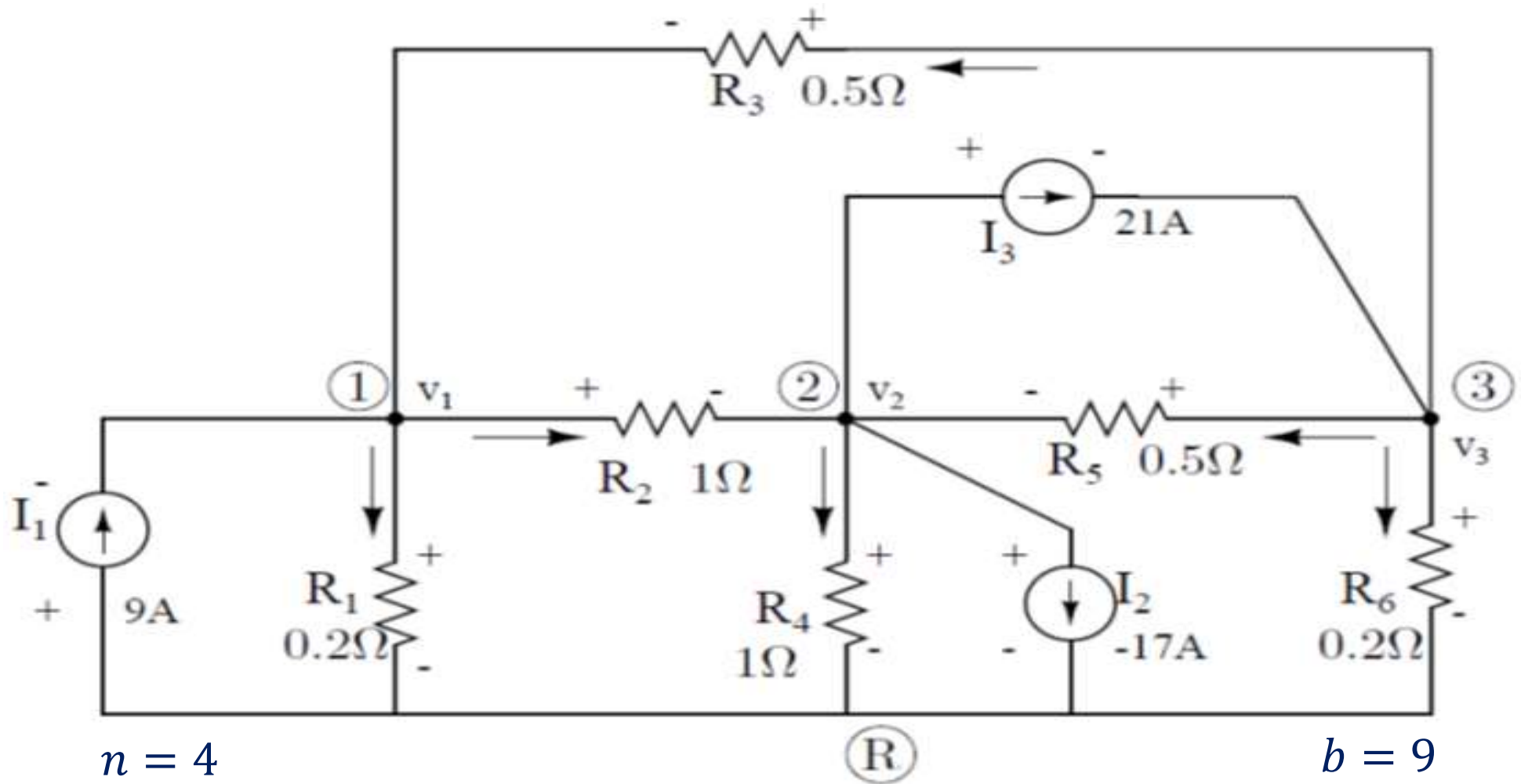
$$V_2 - V_3 + V_4 - V_1 = V_{AB} - V_{CB} + V_{CD} - V_{AD}$$

$$= (V_A - V_B) - (V_C - V_B) + (V_C - V_D) - (V_A - V_D) = 0$$

Circuits with Independent Current Sources



Circuits with Independent Current Sources



Circuits with Independent Current Sources

KCL at Node **1**

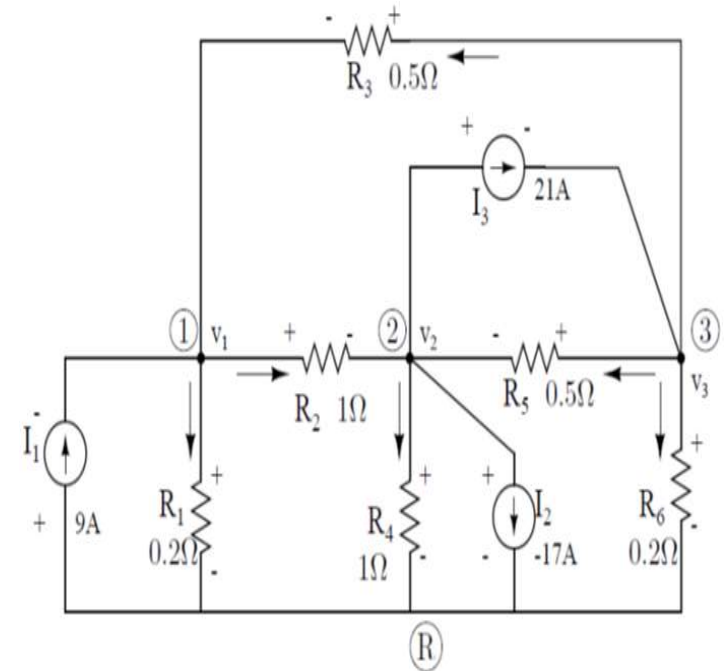
$$-\frac{V_1}{R_1} - \left(\frac{V_1 - V_2}{R_2}\right) + \left(\frac{V_3 - V_1}{R_3}\right) + I_1 = 0$$



$$V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) - V_2 \left(\frac{1}{R_2} \right) - V_3 \left(\frac{1}{R_3} \right) = I_1$$



$$V_1(G_1 + G_2 + G_3) - V_2(G_2) - V_3(G_3) = I_1$$



Circuits with Independent Current Sources

KCL at Node 2

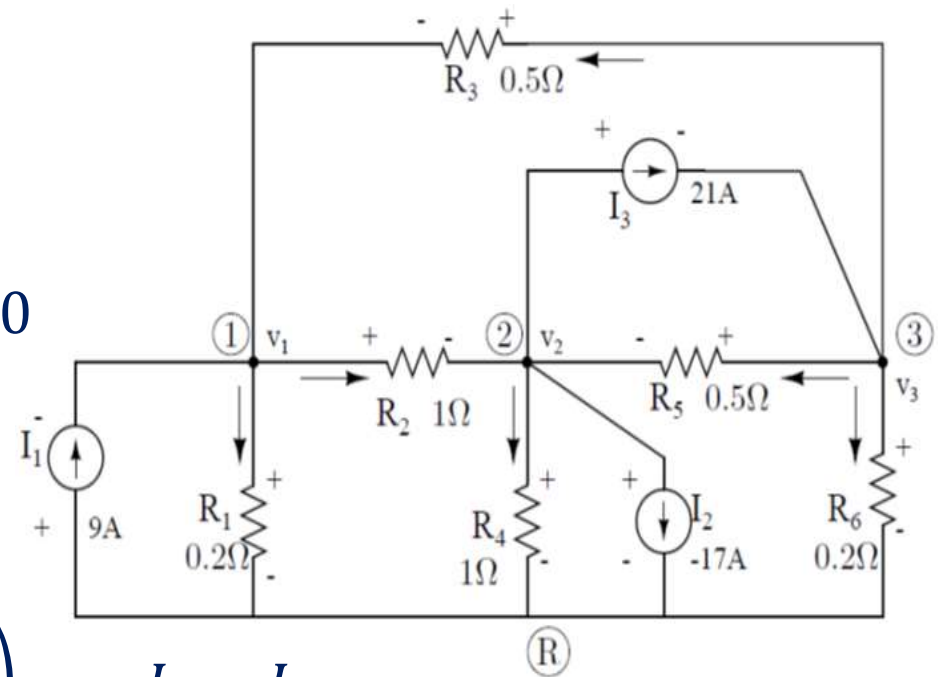
$$\left(\frac{V_1 - V_2}{R_2}\right) - \frac{V_2}{R_4} + \left(\frac{V_3 - V_2}{R_5}\right) - I_2 - I_3 = 0$$



$$-V_1 \left(\frac{1}{R_2}\right) + V_2 \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5}\right) - V_3 \left(\frac{1}{R_5}\right) = -I_2 - I_3$$



$$-V_1(G_2) + V_2(G_2 + G_4 + G_5) - V_3(G_5) = -I_2 - I_3$$



Circuits with Independent Current Sources

KCL at Node 3

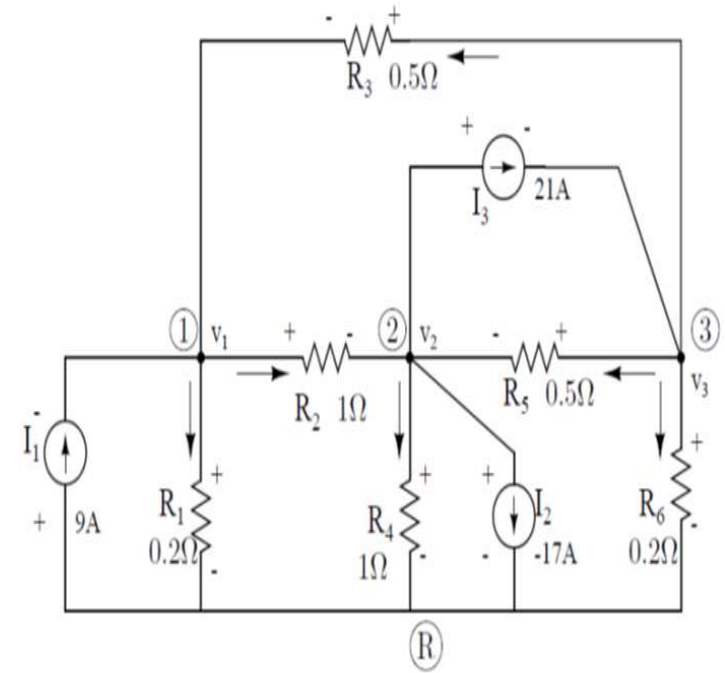
$$-\left(\frac{V_3 - V_2}{R_5}\right) - \left(\frac{V_3 - V_1}{R_3}\right) - \frac{V_3}{R_6} + I_3 = 0$$



$$-V_1 \left(\frac{1}{R_3}\right) - V_2 \left(\frac{1}{R_5}\right) + V_3 \left(\frac{1}{R_3} + \frac{1}{R_5} + \frac{1}{R_6}\right) = I_3$$



$$-V_1(G_3) - V_2(G_5) + V_3(G_3 + G_5 + G_6) = I_3$$

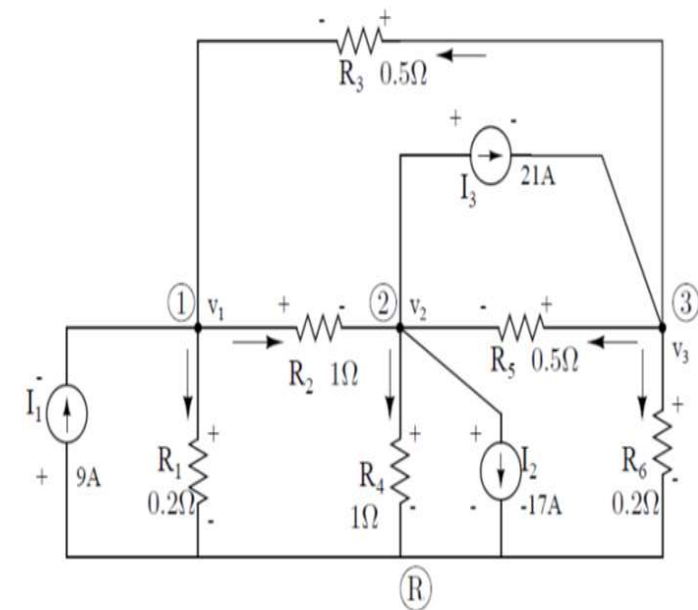


Circuits with Independent Current Sources

$$V_1(G_1 + G_2 + G_3) - V_2(G_2) - V_3(G_3) = I_1$$

$$-V_1(G_2) + V_2(G_2 + G_4 + G_5) - V_3(G_5) = -I_2 - I_3$$

$$-V_1(G_3) - V_2(G_5) + V_3(G_3 + G_5 + G_6) = I_3$$



$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

Circuits with Independent Current Sources

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$



$$\begin{bmatrix} 8 & -1 & -2 \\ -1 & 4 & -2 \\ -2 & -2 & 9 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -4 \\ 21 \end{bmatrix}$$



$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Circuits with Independent Current Sources

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

Conductance Matrix

Input Matrix

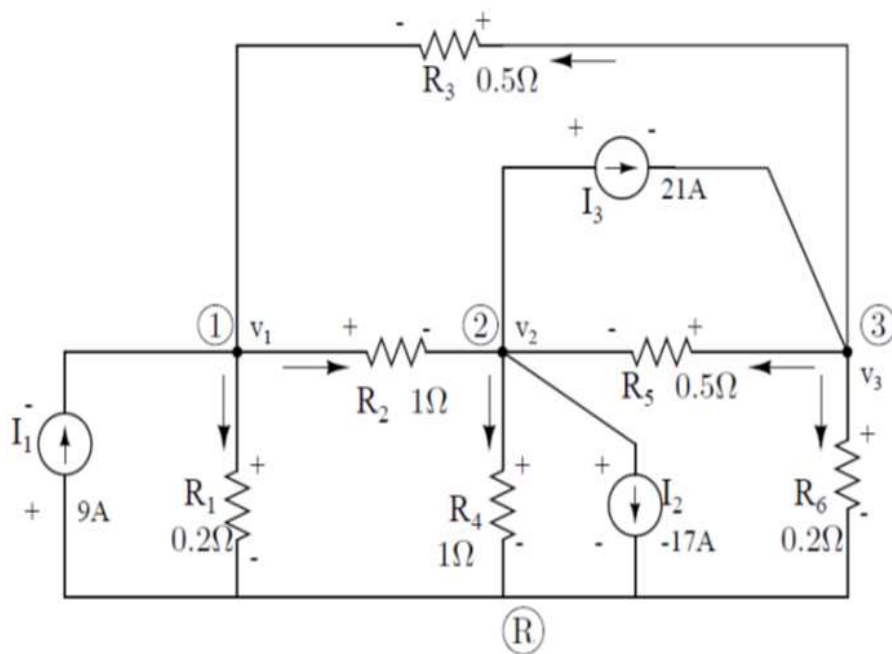
$$[Y]_{(n-1) \times (n-1)} [\underset{\text{Node Voltage Vector}}{V}]_{(n-1) \times 1} = [\underset{\text{Independent Source Vector}}{C}]_{(n-1) \times n_{ics}} [\underset{\text{Independent Source Vector}}{U}]_{n_{ics} \times 1}$$

$\textcolor{blue}{C}\textcolor{red}{U}$: Net Current Injected by Independent Sources at Each Node

n_{ics} : Number of Independent Current Sources

Circuits with Independent Current Sources

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

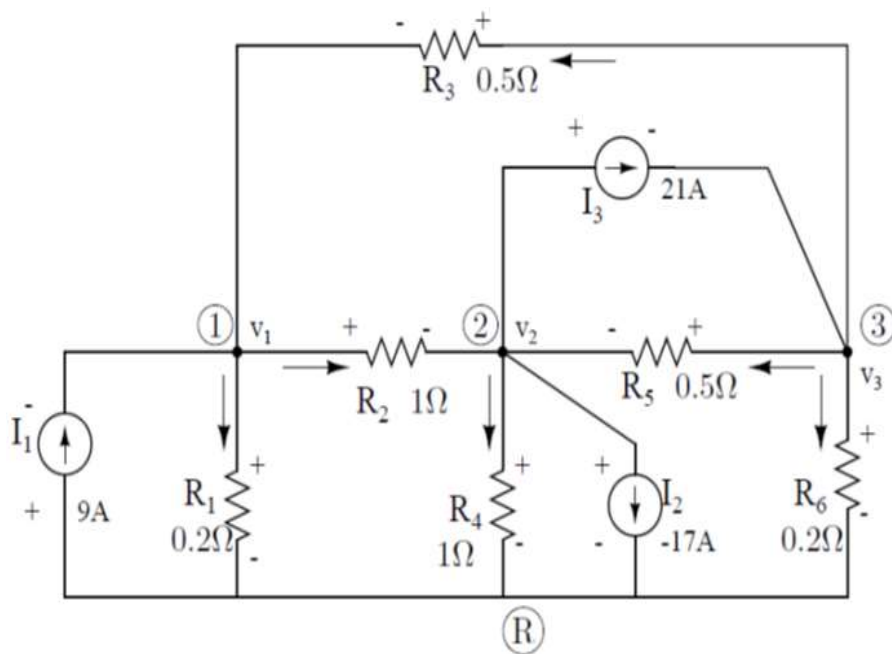


$Y[i][i]$: Sum of All Conductances Connected to Node i

$Y[i][j] = Y[j][i]$: Negative of Sum of All Conductances Connected between Node i and Node j

Circuits with Independent Current Sources

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

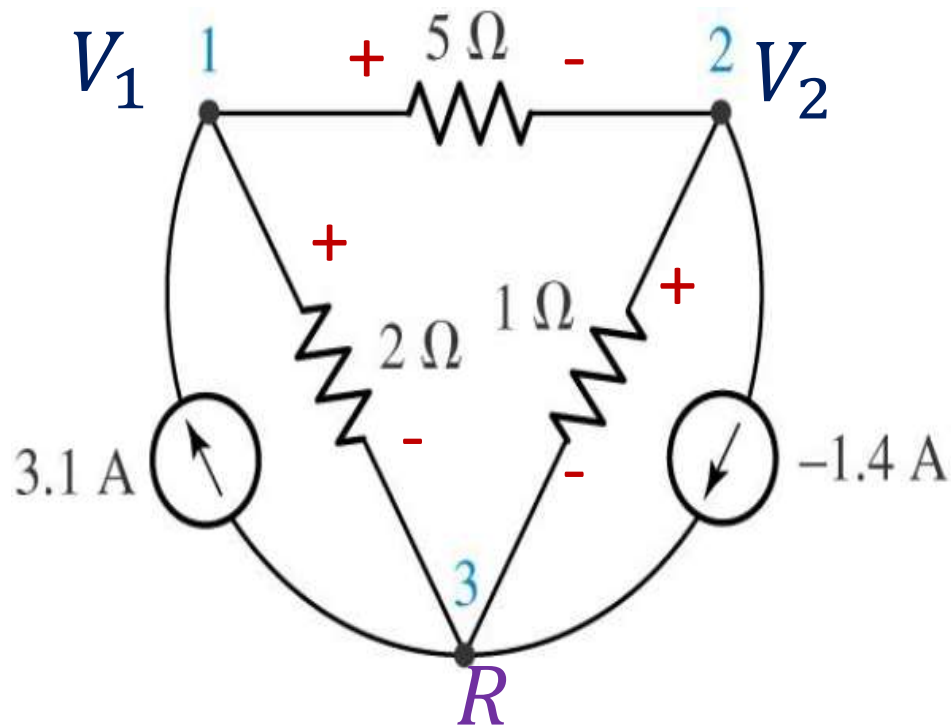


$C[i][j] = 0$: Current Source $U[j]$ Not Connected to Node i

$C[i][j] = 1$: Current Source $U[j]$ is Delivering Current to Node i

$C[i][j] = -1$: Current Source $U[j]$ is Drawing Current from Node i

The Simple Circuit



$$\begin{bmatrix} \frac{1}{2} + \frac{1}{5} & -\frac{1}{5} \\ -\frac{1}{5} & 1 + \frac{1}{5} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3.1 \\ -1.4 \end{bmatrix}$$

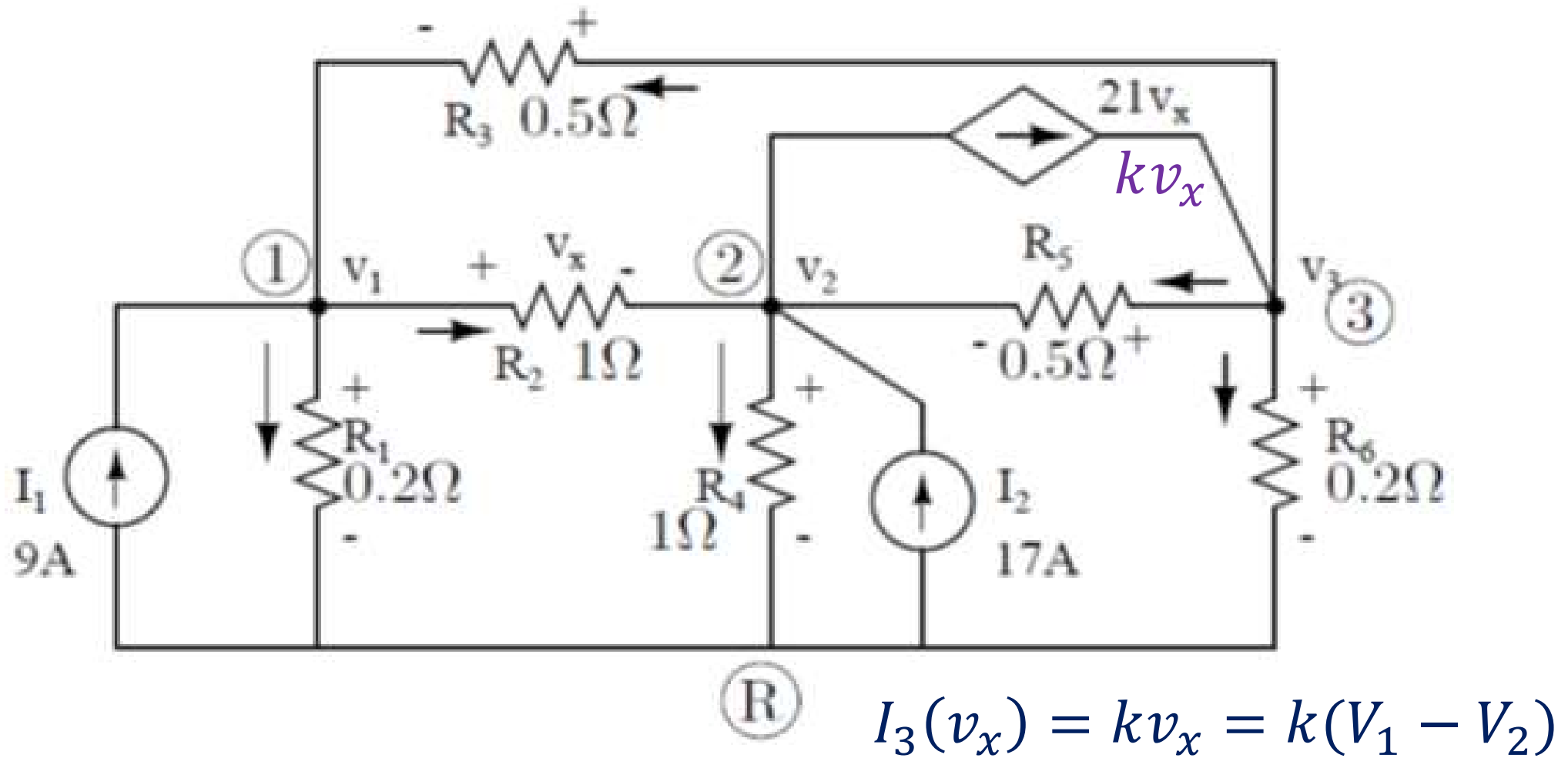


$$\begin{bmatrix} 0.7 & -0.2 \\ -0.2 & 1.2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3.1 \\ -1.4 \end{bmatrix}$$



$$\left. \begin{array}{l} 7V_1 - 2V_2 = 31 \\ V_1 - 6V_2 = -7 \end{array} \right\} \begin{array}{l} V_1 = 5V \\ V_2 = 2V \end{array}$$

Circuits with Dependent Current Sources



Circuits with Dependent Current Sources

KCL at Node **1**

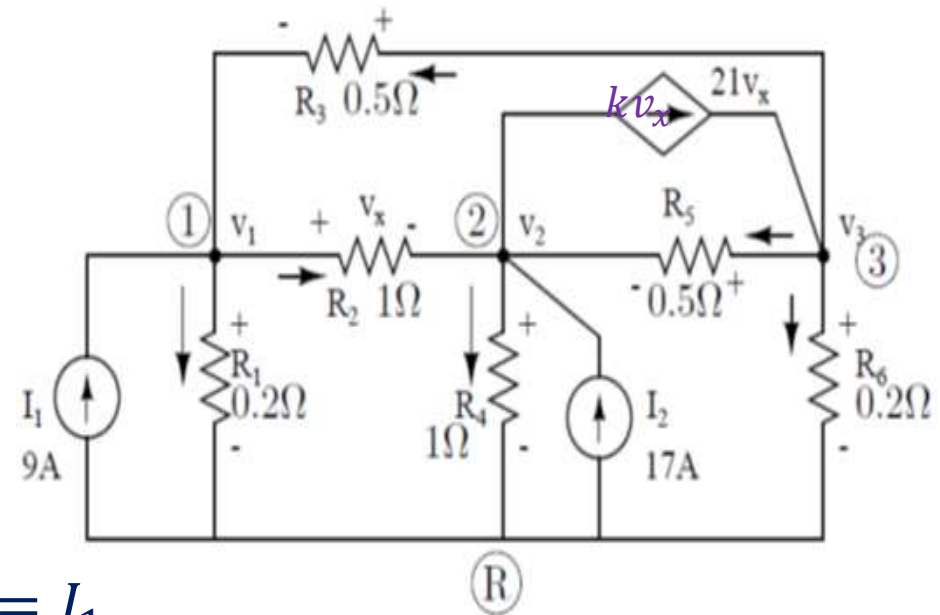
$$-\frac{V_1}{R_1} - \left(\frac{V_1 - V_2}{R_2}\right) + \left(\frac{V_3 - V_1}{R_3}\right) + I_1 = 0$$



$$V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) - V_2 \left(\frac{1}{R_2} \right) - V_3 \left(\frac{1}{R_3} \right) = I_1$$



$$V_1(G_1 + G_2 + G_3) - V_2(G_2) - V_3(G_3) = I_1$$



Circuits with Dependent Current Sources

KCL at Node 2

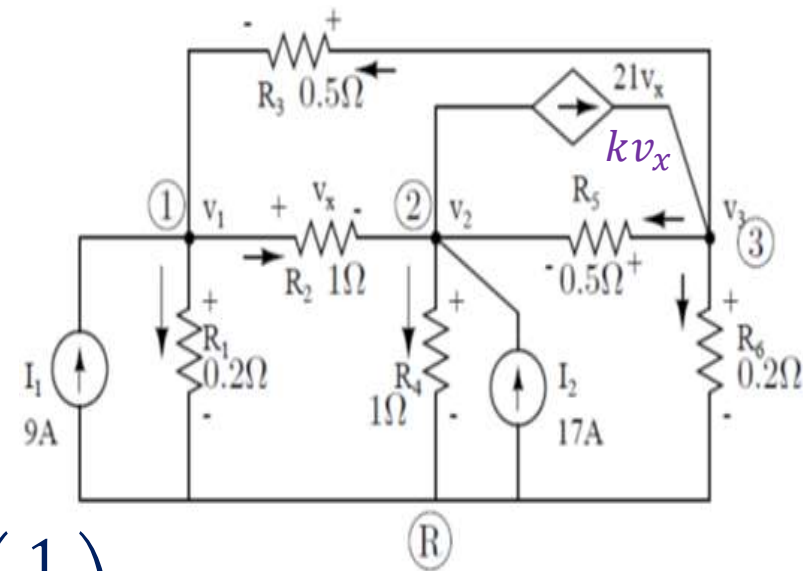
$$\frac{V_1 - V_2}{R_2} - \frac{V_2}{R_4} + I_2 - K(V_1 - V_2) + \frac{V_3 - V_2}{R_5} = 0$$



$$-V_1 \left(\frac{1}{R_2} - K \right) + V_2 \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} - K \right) - V_3 \left(\frac{1}{R_5} \right) = I_2$$



$$-V_1(G_2 - K) + V_2(G_2 + G_4 + G_5 - K) - V_3(G_5) = I_2$$



Circuits with Dependent Current Sources

KCL at Node 3

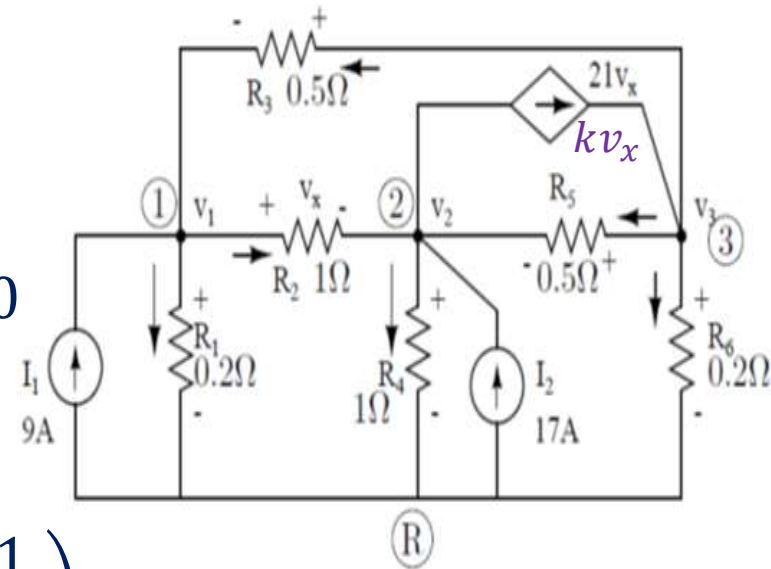
$$-\left(\frac{V_3 - V_1}{R_3}\right) - \left(\frac{V_3 - V_2}{R_5}\right) - \frac{V_3}{R_6} + K(V_1 - V_2) = 0$$



$$-V_1 \left(\frac{1}{R_3} + K \right) - V_2 \left(\frac{1}{R_5} - K \right) + V_3 \left(\frac{1}{R_3} + \frac{1}{R_5} + \frac{1}{R_6} \right) = 0$$



$$-V_1(G_3 + K) - V_2(G_5 - K) + V_3(G_3 + G_5 + G_6) = 0$$

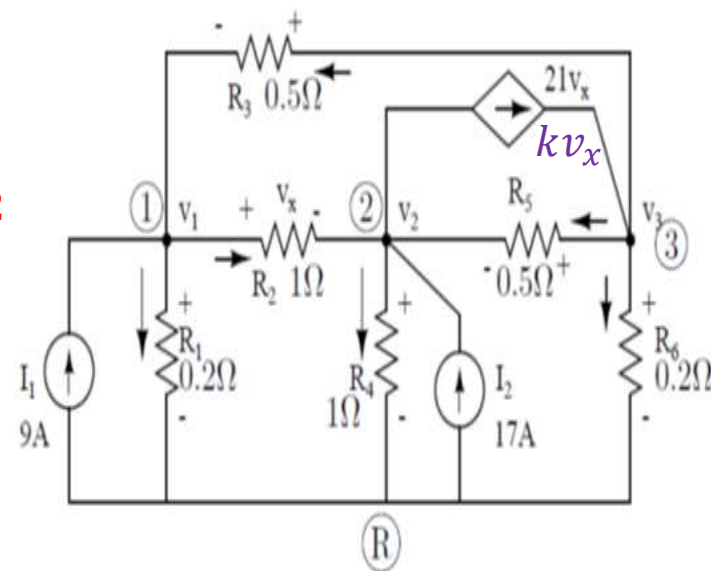


Circuits with Dependent Current Sources

$$V_1(G_1 + G_2 + G_3) - V_2(G_2) - V_3(G_3) = I_1$$

$$-V_1(G_2 - K) + V_2(G_2 + G_4 + G_5 - K) - V_3(G_5) = I_2$$

$$-V_1(G_3 + K) - V_2(G_5 - K) + V_3(G_3 + G_5 + G_6) = 0$$



$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -(G_2 - K) & G_2 + G_4 + G_5 - K & -G_5 \\ -(G_3 + K) & -(G_5 - K) & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Circuits with Dependent Current Sources

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -(G_2 - K) & G_2 + G_4 + G_5 - K & -G_5 \\ -(G_3 + K) & -(G_5 - K) & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$



$$\begin{bmatrix} 8 & -1 & -2 \\ 20 & -17 & -2 \\ -23 & 19 & 9 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 9 \\ 17 \end{bmatrix}$$



$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Summary

- Introduction to Nodal Analysis
- Evaluating Node Voltages w.r.t. a Reference Node
- Circuits with Independent Current Sources
- The Nodal Analysis Equation: $YV = CU$
- Circuits with Dependent Current Sources



Thank You